

MATH 127 Calculus for the Sciences

Lecture 23



Today's lecture

Last time

How to solve indeterminate forms $\frac{\infty}{\infty}, \frac{0}{0}$.

This time

Course note coverage Section 3.2.4 and 3.2.5

Indeterminate product and indeterminate powers.



Happy Halloween

I have chocolate for you. Here is a picture of me visiting the Lindt chocolate factory to buy you this great chocolate.



Lindt  HOME OF CHOCOLATE



Recap

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
$\frac{1}{0^-}, \frac{\infty}{0^-}$	$-\infty$
$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0
Indeterminate form	tricks
$\frac{\infty}{\infty}, \frac{0}{0}$	L'Hospital's rule simplify multiply conjugate
$0 \cdot \infty$ (new!)	will see today
$0^0, \infty^0, 1^\infty$ (new!)	will see today



Indeterminate product

Question What do we do with a indeterminate form like $0 \cdot \infty$?

We could rewrite

$$0 \cdot \infty = \frac{0}{(1/\infty)} = \boxed{\frac{0}{0}}$$

$$\begin{aligned} A \cdot B &= A / (1/B) \\ &= B / (1/A) \end{aligned}$$

And we know how to deal with $\frac{0}{0}$ from before!

Alternatively...

$$0 \cdot \infty = \frac{\infty}{(1/0)} = \boxed{\frac{\infty}{\infty}}$$

And we know how to deal with $\frac{\infty}{\infty}$ from before!



Example: Indeterminate product

Example

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} \right) \cdot \sin(x) \rightarrow +\infty \cdot 0^+$$

The limit of $\frac{1}{x}$ as $x \rightarrow 0^+$ is

$$+\infty$$

The limit of $\sin(x)$ as $x \rightarrow 0^+$ is

$$0^+$$

So this limit has indeterminate form

$$\infty \cdot 0$$

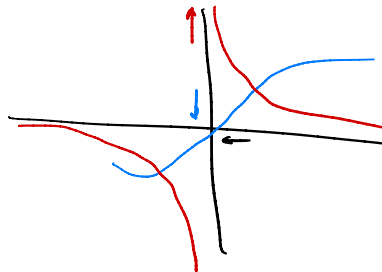
But we could rewrite the limit as a quotient

$$\lim_{x \rightarrow 0^+} \frac{\sin(x)}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{\sin(x)}{x} = \lim_{x \rightarrow 0^+} \frac{\cos x}{1} = 1$$

This has indeterminate form

$$\frac{0^+}{0^+}$$

which can be solved using **L'Hospital** rule.





Example: Indeterminate product

Example

$$\lim_{x \rightarrow \infty} x \cdot \sin\left(\frac{1}{x}\right)$$

The limit of x as $x \rightarrow \infty$ is

$$\infty$$

The limit of $\sin\left(\frac{1}{x}\right)$ as $x \rightarrow \infty$ is

$$0^+$$

So this limit has indeterminate form

$$\infty \cdot 0^+$$

But we could rewrite the limit as a quotient

$$\lim_{x \rightarrow \infty} \frac{x}{1/\sin\left(\frac{1}{x}\right)}, \quad \text{or} \quad \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{1/x}$$

These have indeterminate forms

$$\frac{\infty}{\infty} \quad \text{and} \quad \frac{0}{0}$$

respectively... but which one should we use?



Example: Indeterminate product

Example

$$\lim_{x \rightarrow \infty} \frac{x}{1/\sin\left(\frac{1}{x}\right)},$$

or

$$\lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{1/x}$$

These have indeterminate forms

$$\frac{\infty}{\infty} \text{ and } \frac{0}{0}$$

respectively... but which one should we use?

We are planning to apply L'Hôpital's rule. So we need to take derivatives of the denominator and numerator. Which one is easier?

$$\lim_{x \rightarrow \infty} \frac{\sin(\frac{1}{x})}{1/x} = \lim_{x \rightarrow \infty} \frac{\cos(\frac{1}{x})(\frac{1}{x})'}{(\frac{1}{x})'} = \lim_{x \rightarrow \infty} \cos(\frac{1}{x}) = 1$$

Since the right hand side exists, we can apply L'Hospital's rule.



Example: Indeterminate product

Example

$$\lim_{x \rightarrow \infty} \frac{x}{1/\sin\left(\frac{1}{x}\right)}, \quad \text{or} \quad \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{1/x}$$

The **second one** is easier to apply L'Hospital's rule, because who on earth would want to take the derivative of $1/\sin(1/x)$?

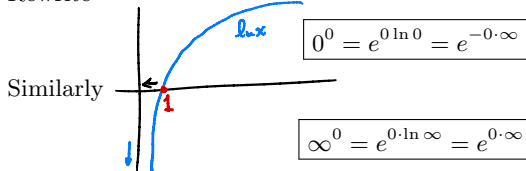


Indeterminate power

Question What do we do with a indeterminate form like 0^0 , ∞^0 , 1^∞ ?

Did you forget what I said on Sept 24? Always use base e !

Rewrite



and

$1^\infty = e^{\infty \cdot \ln 1} = e^{\infty \cdot 0}$

$$\begin{aligned}
 (x)^0 &= 1 \leftarrow 0^0 \\
 0^{(x)} &= 0 \leftarrow 0^0 \\
 (x)^0 &= 1 \leftarrow \infty^0 \\
 \infty^{(x)} &= \infty \leftarrow \infty^0 \\
 1^{(x)} &= 1 \leftarrow 1^\infty \\
 (x)^\infty &= \begin{cases} \infty, & (x) > 1 \\ 0, & 0 < (x) < 1 \end{cases} \leftarrow 1^\infty
 \end{aligned}$$

Now we just have to solve for the $0 \cdot \infty$ thing (which we just saw how to do) and take e to the power of it.



Example: indeterminate power

Question Evaluate $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$.

This is the indeterminate form ∞^0

$$\lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \lim_{x \rightarrow \infty} e^{\frac{1}{x} \ln x}$$

$$= e^{\lim_{x \rightarrow \infty} \frac{1}{x} \ln x}$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} \ln x = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

by continuity of e^x

$$= \exp\left(\lim_{x \rightarrow \infty} \frac{\ln x}{x}\right)$$

The new limit is
of form $0/\infty$.

$$= \exp\left(\lim_{x \rightarrow \infty} \frac{\ln x}{x}\right)$$

The new limit is of form $\frac{\infty}{\infty}$

$$= \exp\left(\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1}\right) = \exp\left(\lim_{x \rightarrow \infty} \frac{1}{x}\right) = \exp(0) = e^0 = 1$$

The RHS indeed exists
so L'Hospital's rule applies



Example: indeterminate power

Question Evaluate $\lim_{x \rightarrow 0} \sin(x)^{\tan(x)}$.

This is indeterminate form 0^0

$$\lim_{x \rightarrow 0} \sin(x)^{\tan(x)} = \lim_{x \rightarrow 0} e^{\tan(x) \ln(\sin(x))}$$

$$= \exp\left(\lim_{x \rightarrow 0} \underbrace{\tan(x)}_{\uparrow} \cdot \underbrace{\ln(\sin(x))}_{\uparrow}\right)$$

This limit has indeterminate form $0 \cdot (-\infty)$.

$$= \exp\left(\lim_{x \rightarrow 0} \frac{\ln(\sin(x))}{1/(\tan x)}\right)$$

This limit has form $\frac{-\infty}{\pm \infty}$

$$= \exp\left(\lim_{x \rightarrow 0} \frac{\frac{1}{\sin(x)} \cdot \cos x}{\frac{-1}{(\tan(x))^2} \cdot \sec^2(x)}\right) = \exp\left(\lim_{x \rightarrow 0} \frac{\cos^3(x) \cdot (-\tan^2(x))}{\sin(x)}\right)$$

$$\exp\left(\lim_{x \rightarrow 0} \frac{-\cos^3(x) \cdot \boxed{\tan^2(x)}}{\sin(x)}\right)$$

$$= \exp\left(\lim_{x \rightarrow 0} \frac{-\boxed{\cos^3(x)} \cdot \frac{\boxed{\sin^2(x)}}{\boxed{\cos^2(x)}}}{\boxed{\sin(x)}}\right)$$

$$= \exp\left(\lim_{x \rightarrow 0} -\cos(x) \sin(x)\right)$$

$$= \exp\left(-\underbrace{\lim_{x \rightarrow 0} \cos(x)}_1 \cdot \underbrace{\lim_{x \rightarrow 0} \sin(x)}_0\right)$$

$$= \exp(0) = 1$$

Since RHS exists, it was valid to apply the L'Hospital's rule



Recap

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$\frac{\infty}{\infty}, \frac{0}{0}$	L'Hospital's rule simplify multiply conjugate
$0 \cdot \infty$	rewrite as a quotient and apply above
$0^0, \infty^0, 1^\infty$	rewrite in base e and apply above